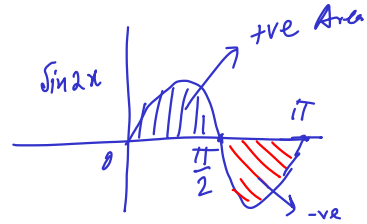
Tutorial - 3

Q1 (a) Find the area under the curve $y = f(x) = \sin 2x$ on $[0, \frac{\pi}{2}]$

(b) Compute the integral $\int \sin 2x \, dx$ for interval $(0, \pi)$. Is this double of answer in (a)?

$$A_1 = \int_0^{\pi/2} f(x)\, dx = \int_0^{\pi/2} \sin 2x \, dx = \frac{1}{2}\left[-\cos(\pi) + \cos 0\right]$$

$$= \frac{1}{2}(1+1) = 1$$



$$\frac{1}{2}\frac{d}{dx}(-\cos 2x) = \sin 2x$$

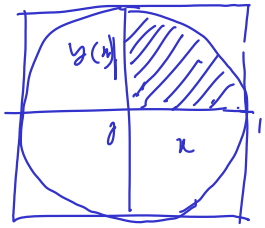
$$A_2 = \int_0^{\pi} \sin 2x \, dx = \int_0^{\pi/2} \sin 2x \, dx + \int_{\pi/2}^{\pi} \sin 2x \, dx = 1 + \frac{1}{2}\left[-\cos(2\pi) + \cos \pi\right]$$

$$= 1 - 1 = 0$$

Clearly $A_2 \neq 2A_1$, as area between $(\pi, 2\pi)$ contributes negatively

Q2 Compute the area encircled by a unit circle in the first quadrant.

Eg. 2:



unit circle $x^2 + y^2 = 1$ $\quad y = \sqrt{1-x^2}$

$$\text{Shaded Area} = \int_0^1 y\, dx = \int_0^1 \sqrt{1-x^2}\, dx$$

Say, $x = \sin u$, $dx = \cos u \, du$

$2\cos^2 u = \cos 2A + 1$

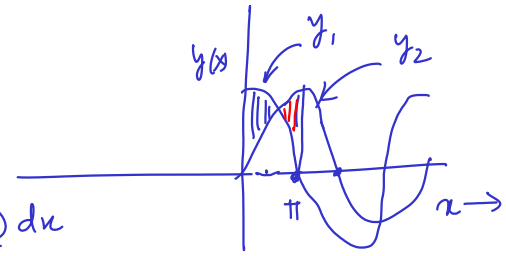
$$= \int_0^{\pi/2} \cos^2 u \, du = \frac{1}{2}\int_0^{\pi/2} \cos 2A \, dx + \frac{1}{2}\int_0^{\pi/2} dx$$

$$= \left[\sin \pi - \sin 0\right] + \frac{1}{2}\left[\frac{\pi}{2} - 0\right] = \frac{\pi}{4}$$

Q3 (a) Find the area between the curve $y_1(x) = \sin\frac{x}{2}$ and $y_2(x) = \cos\frac{x}{2}$ between $[0, \pi/2]$

(b) What would be the area between the curve on $[0, \pi]$

(a)



The area between the curve

$$A_1 = \int_0^{\pi/2} \left(y_2(x) - y_1(x)\right) dx = \int_0^{\pi/2} \left(\cos\frac{x}{2} - \sin\frac{x}{2}\right) dx$$

$$= 2\left[+\sin x\Big|_0^{\pi/4} + \cos x\Big|_0^{\pi/4}\right] = 2\left[\sin\frac{\pi}{4} - \sin(0) + \cos\frac{\pi}{4} - \cos 0\right] = 2\left[\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - 1\right]$$

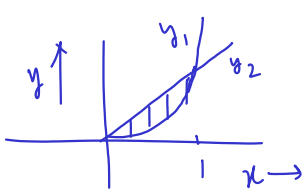
$$\boxed{A_1 = 2(\sqrt{2} - 1)}$$

(b) We note that for $x \in (0, \pi/2)$ $\cos\frac{x}{2} > \sin\frac{x}{2}$ but for $x \in (\frac{\pi}{2}, \frac{\pi}{4})$ $\sin\frac{x}{2} > \cos\frac{x}{2}$

$$\text{Area} = A_2 = \int_0^{\pi/2} \left(\cos\frac{x}{2} - \sin\frac{x}{2}\right) dx + \int_{\pi/2}^{\pi} \left(\sin\frac{x}{2} - \cos\frac{x}{2}\right) dx$$

$$= 2(\sqrt{2}-1) + 2(\sqrt{2}-1) = 4(\sqrt{2}-1)$$

Q4 Find the area between the curve $y_1(x) = x^2$ and $y_2(x) = x$ over $[0,1]$.



For $x \in [0,1]$ $\quad y_1 \leq y_2$

$$\text{Area} = \int_0^1 \left[y_2(x) - y_1(x)\right] dx = \int_0^1 \left[x - x^2\right] dx$$

$$= \frac{x^2}{2}\Big|_0^1 - \frac{x^3}{3}\Big|_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$